

Semiconductor Fabs

Overview + Q&A

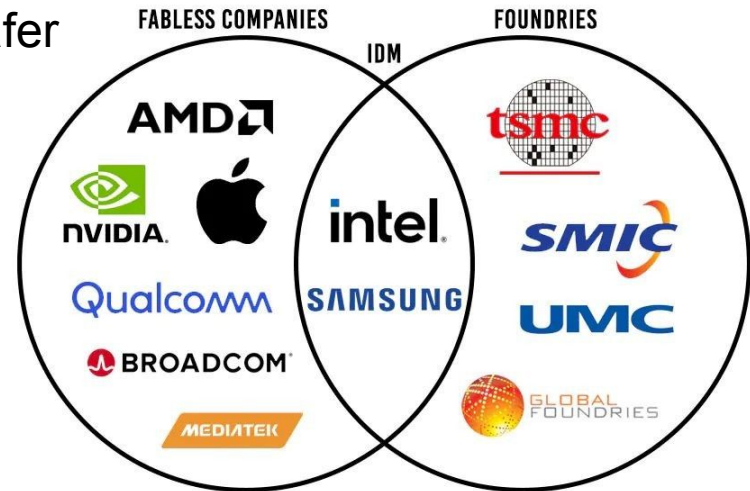
No Magic Pill

Background

- Myself:
 - Process and equipment engineer at 300 mm fab
 - I don't know everything, but I especially don't know chip design—manufacturing only!
- Presentation:
 - Similar to LessOnline 2025, but with some changes
 - Some images taken from Construction Physics' How to Build a \$20 Billion Semiconductor Fab
 - Feel free to ask questions throughout
 - This meant as overview with Q&A at the end

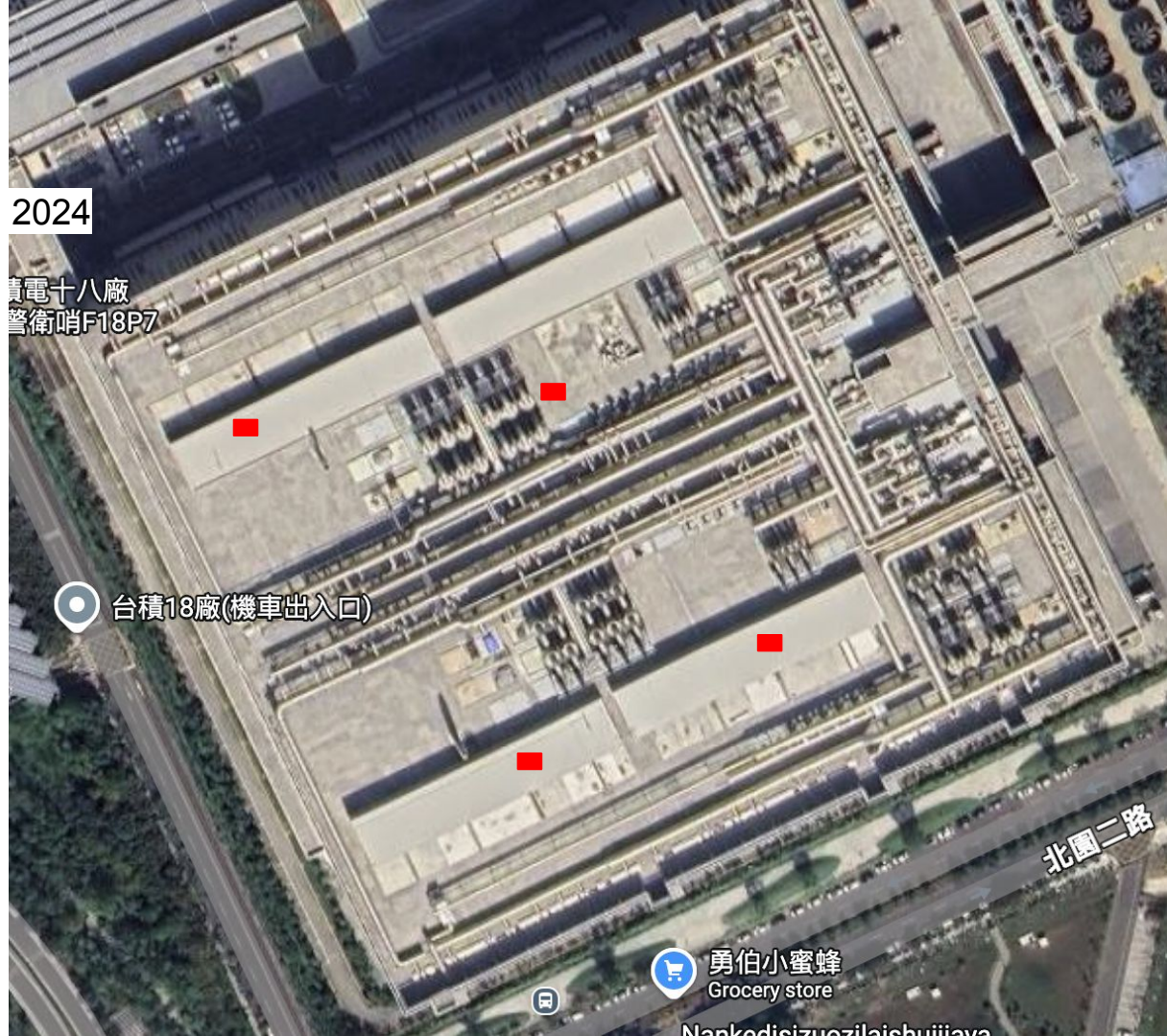
What is a Fab?

- Short for semiconductor **fab**rication facility
- Factory where electronic chips are made on silicon discs (wafers)
- Houses 100s to 1000s of machines that process the chips
- \$5-20B just to build a modern fab
- 500-3000+ wafers/day @ 100-100k chips/wafer
- Fab and fabless companies



Scale! It's Crazy!

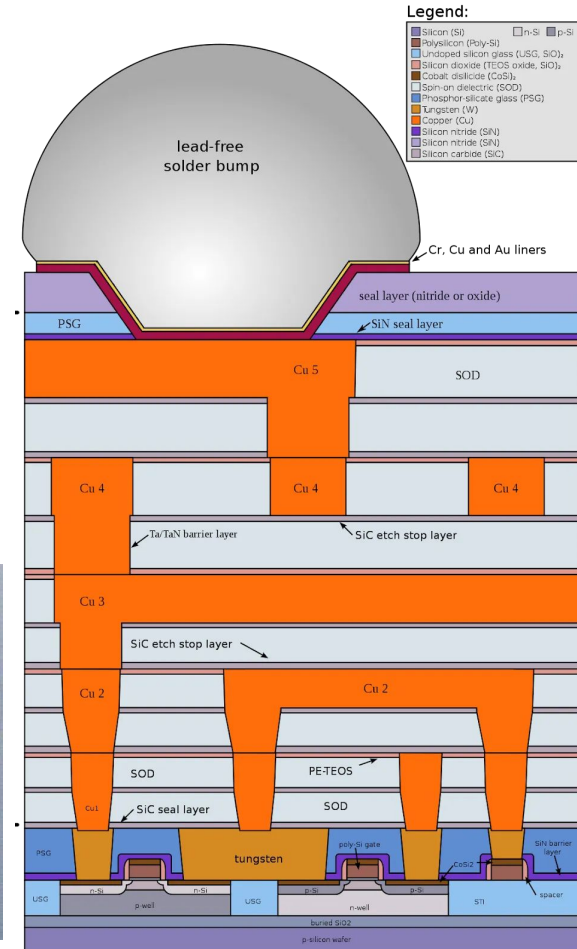
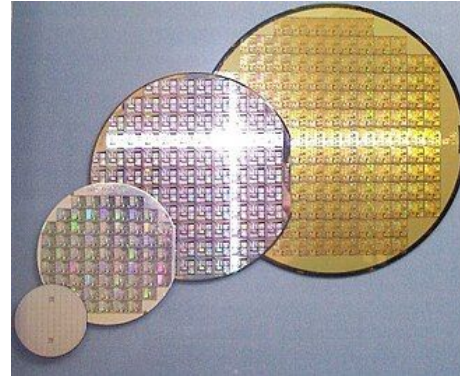
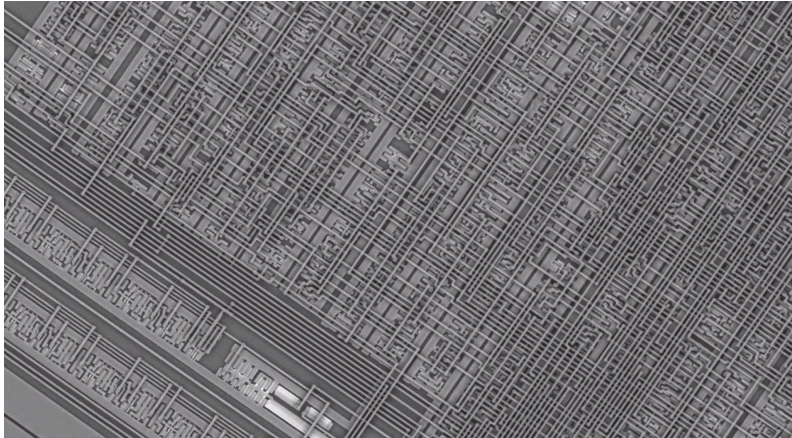
- TSMC F18 put out 2M wafers in 2024
- Red blocks are ~2 cars of area





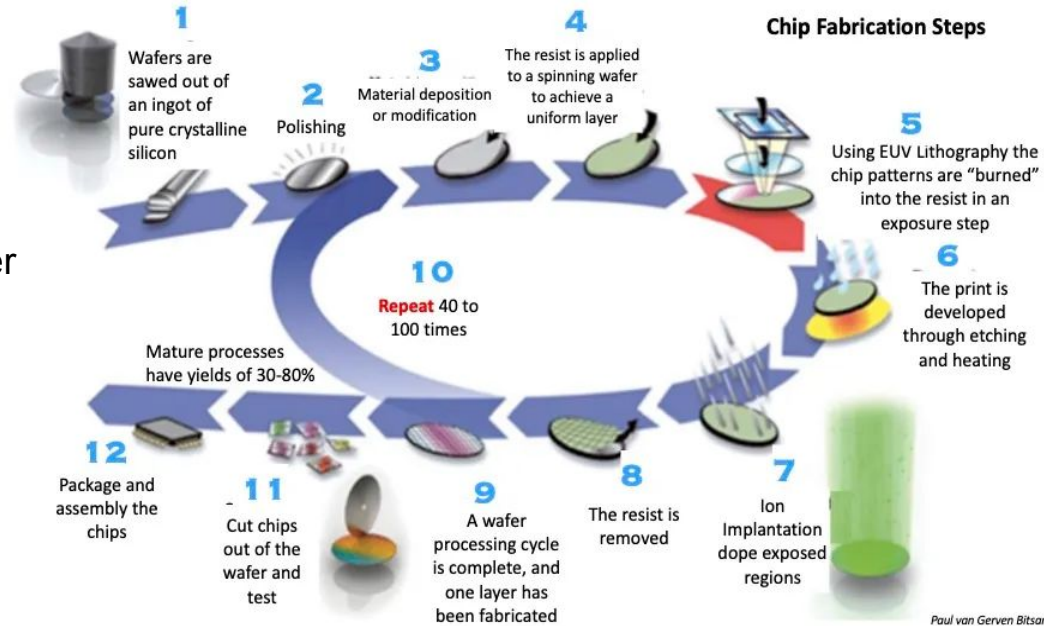
What Are We Building in a Fab?

- A lot of transistors at the surface of the wafer
- Even more interconnects on top of the transistors
 - Interconnects allow chip to interface with the outside world
- Incredibly small at nanometer scale!
 - Human hair is 100,000 nm



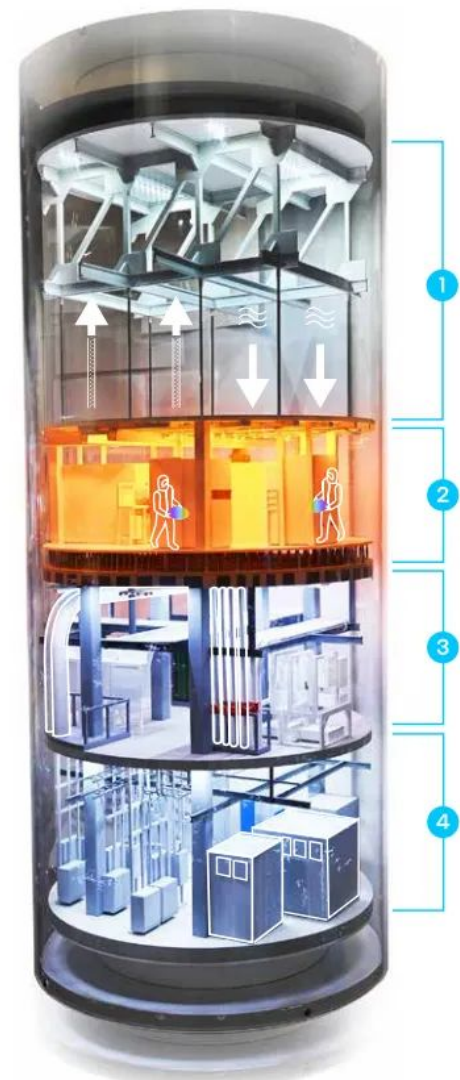
Core Processes for Building a Chip

- Additive: deposition or growth
 - Ex: Depositing a metal layer
- Subtractive: etching
 - Ex: Removing certain parts of a layer
- Modification: anneal or implantation
 - Ex: Implanting ions into wafer
- Patterning: lithography
 - This is what ASML specializes in
 - Allows for “blanket” processes



The Fab Stack

- Foundation is critical to minimize vibrations
 - Remember the nm scale we are working on here!
 - Lithography tools go in middle of fab w/ special vibration dampers
- Typically four levels:
 - Above cleanroom: Fans, filters
 - Cleanroom: Equipment (also called tools)
 - Subfab 1: Facilities (power, gases, chillers)
 - Up to 50 gases (some very toxic), miles of piping
 - Subfab 2: Facilities (power, water, exhaust)

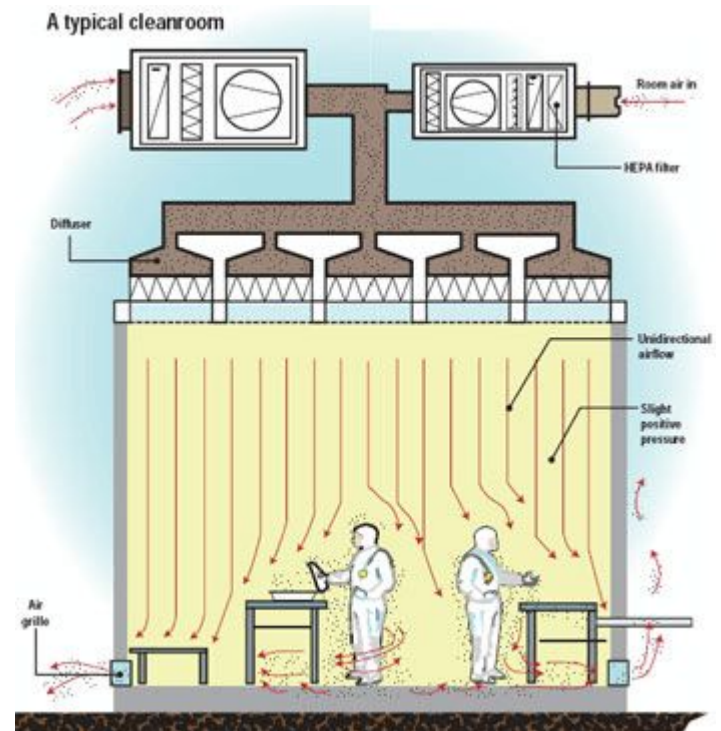


Cleanrooms: The Cleanest Places on Earth

- Air is pushed from top down and recirculates from sides back up
- More circulation + more filters = cleaner
- Inside of tools is even cleaner (mini envs)
- Humidity, temp, and pressure controlled

ISO 14644-1 Cleanroom Standards | Cleanroom Classifications

Class	Maximum Particles/m ³						FED STD 209E equivalent
	≥0.1 um	≥0.2 um	≥0.3 um	≥0.5 um	≥1 um	≥5 um	
ISO 1	10	2					
ISO 2	100	24	10	4			
ISO 3	1,000	237	102	35	8		Class 1
ISO 4	10,000	2,370	1,020	352	83		Class 10
ISO 5	100,000	23,700	10,200	3,520	832	29	Class 100
ISO 6	1,000,000	237,000	102,000	35,200	8,320	293	Class 1,000
ISO 7				352,000	83,200	2,930	Class 10,000
ISO 8	←			3,520,000	832,000	29,300	Class 100,000

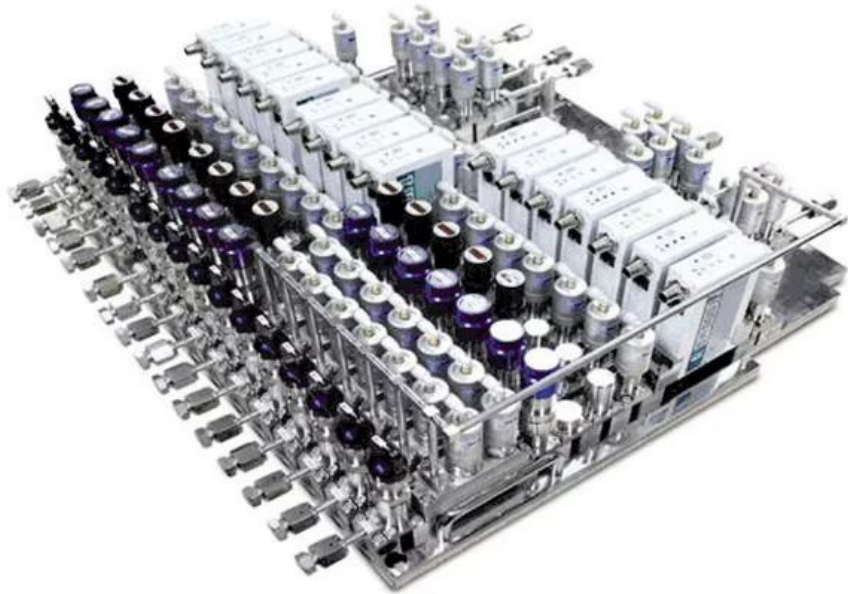


Facilities - Power

- Fabs use ~100 MWh/hr, or 100,000 kWh/hr
 - Avg home uses 1.5 kWh/hr, ~60000x less than fab
- Backup generators onsite in case of power outage
- Companies get energy supply guarantees before building

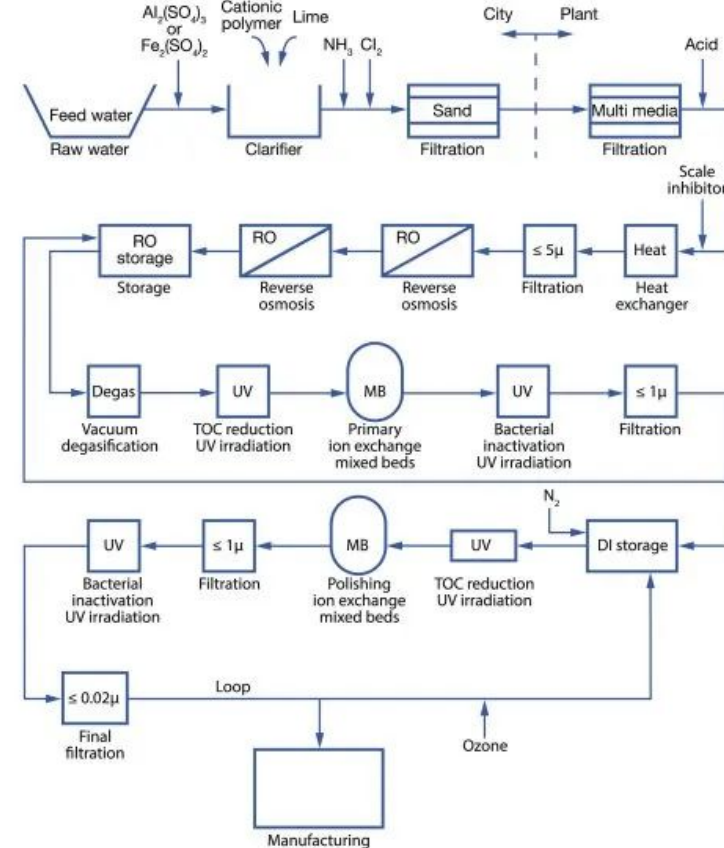
Facilities - Gas

- Up to 50 unique gases, many of which are toxic
- Ranges from cheap (nitrogen) to expensive (helium, NF₃)
- Miles of piping to deliver from gas supply to tool



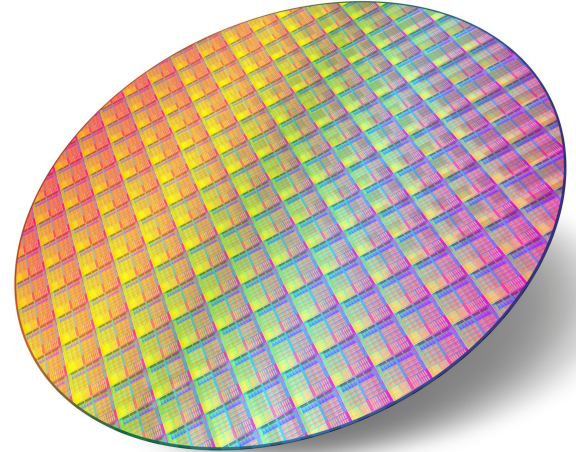
Facilities - Water

- Avg fab uses 10 mil gallons of water per day!
- Recycling rates very high at 65-95%
 - Important for places with limited water (Arizona)
- Used to cool machines or processing



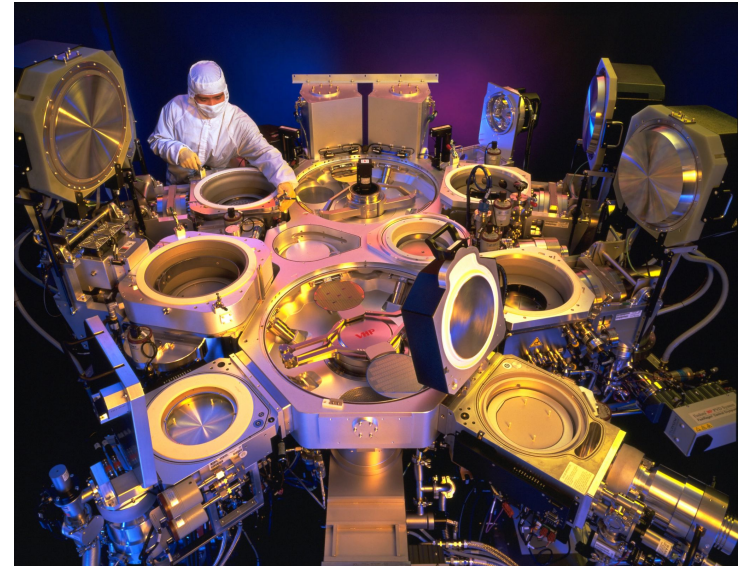
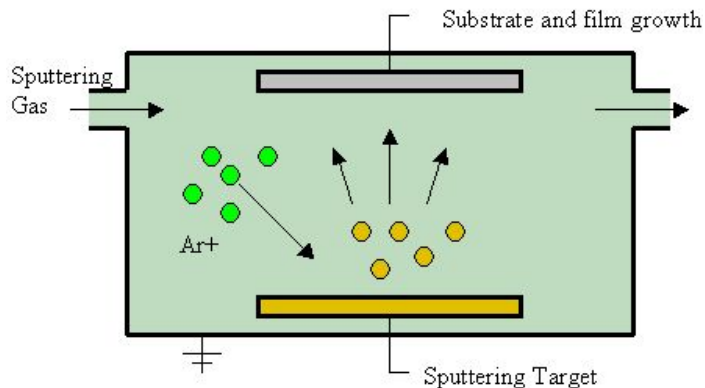
Why Silicon?

- Super cheap and plentiful
- Silicon crystals are ultrapure (99.9999999%)
- We can easily control electrical properties (doping)
- Silicon ingots are cut into wafers
 - Wafers are what the chips are built on
 - 300 mm diameter wafer is current standard



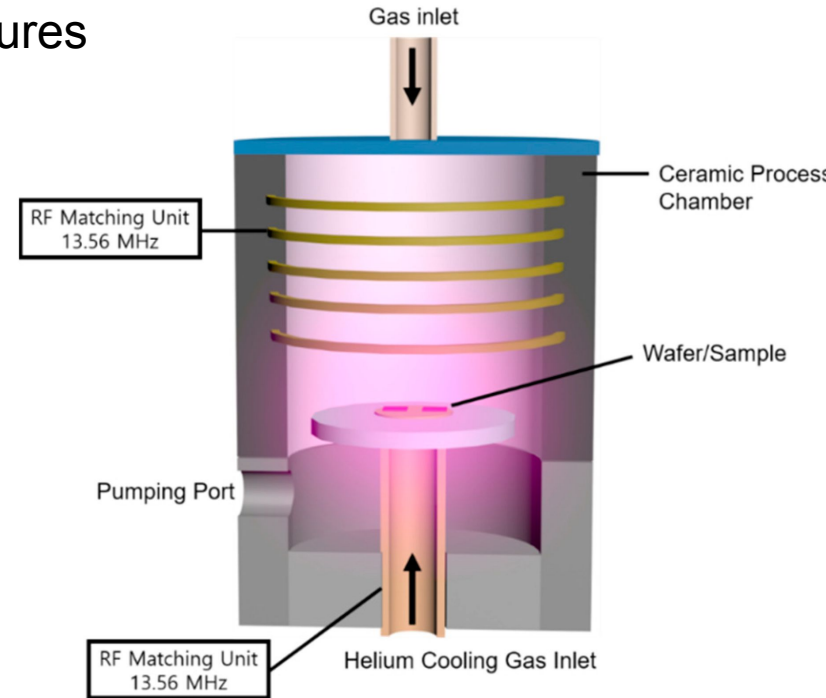
Additive Processes

- Physical vapor deposition (PVD)
 - Slams ions into giant piece of metal to eject atoms, which then get onto wafer
- Chemical vapor deposition (CVD)
 - Gases react to form film on the surface of the wafer
- Oxidation
 - Gases react with surface of wafer to form SiO_2



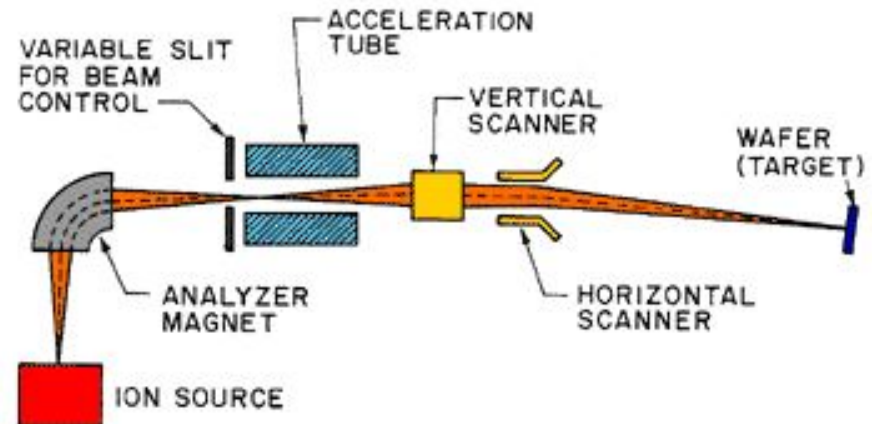
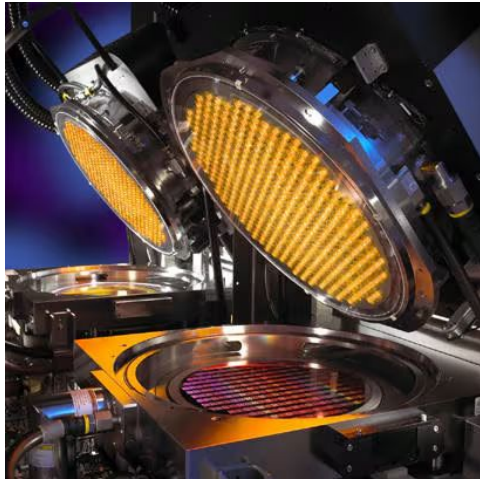
Subtractive Processes (Etching)

- Two types: dry (plasma) and wet (liquid chemicals, like acids)
- Selectivity (if process only removes what we want removed) matters!
- Dry etching good for small and vertical features
- Wet etching good for high selectivity

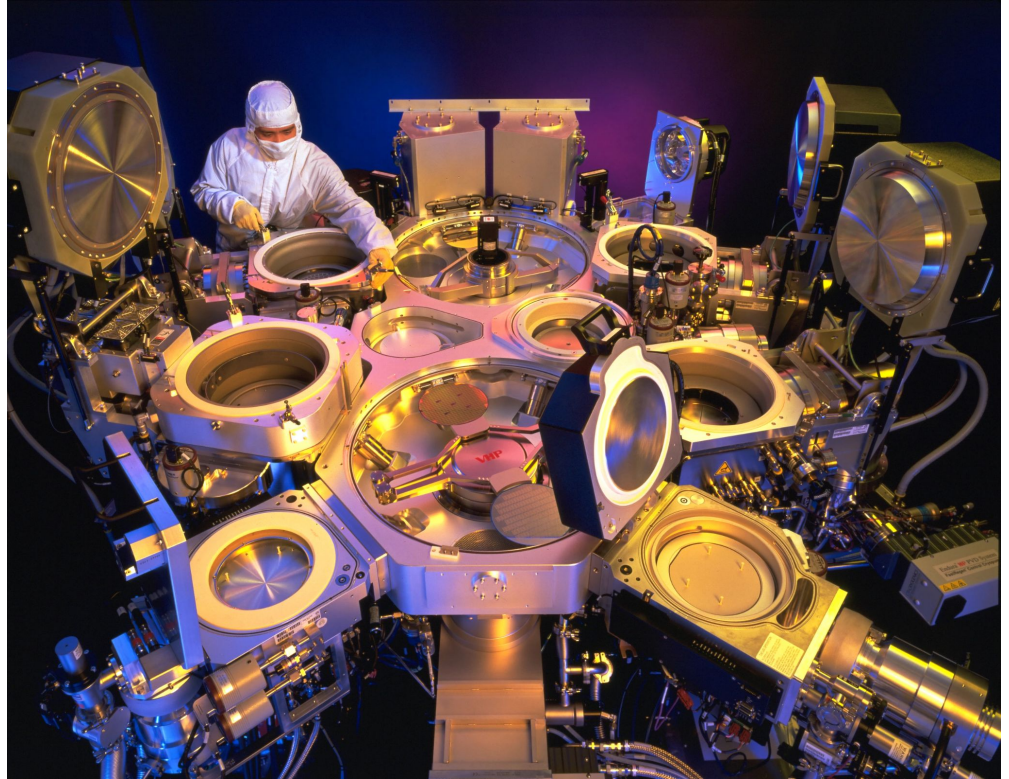
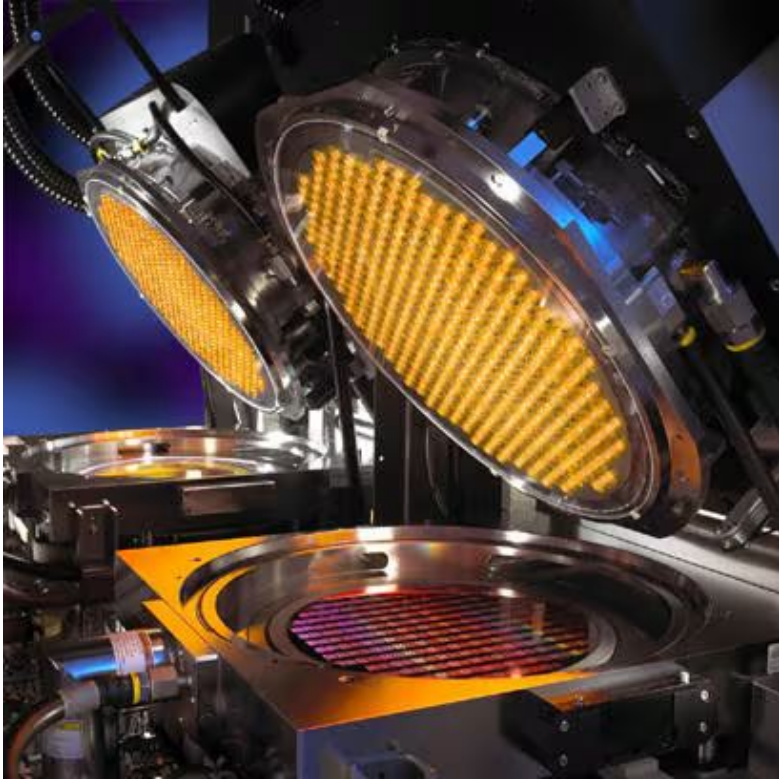


Modification Processes

- Implantation:
 - Giant magnets accelerate ions into wafers to change electrical properties (doping)
- Annealing:
 - Heats the wafer up to high temps to “lock in” recent modifications
 - Temps between 900-1200 °C

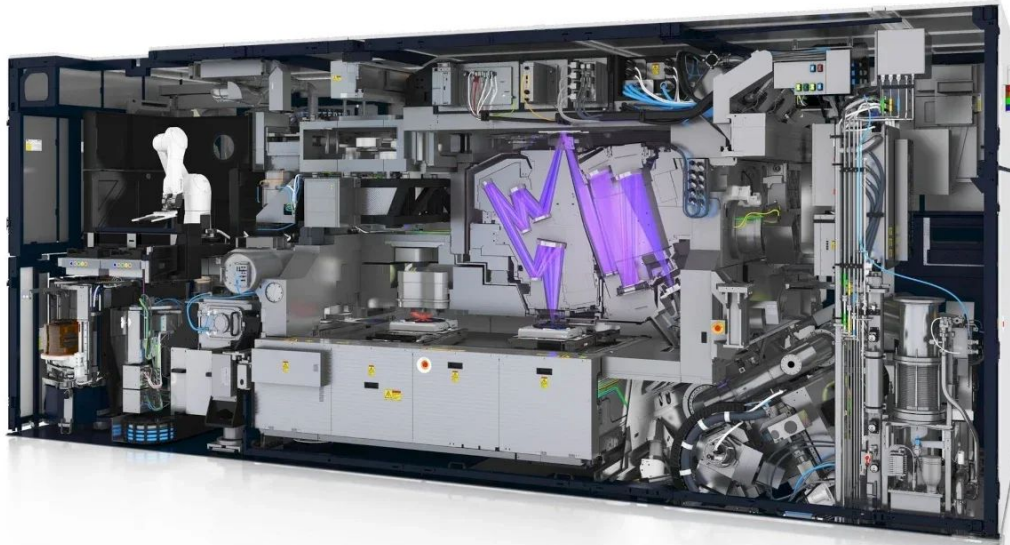


What Does a “Tool” Even Look Like?



Lithography

- Uses light and photomask to expose certain parts of the wafer
- The “bread and butter” of semi industry
- Most complex machines ever built (sorry, NASA!)
- Tools are extremely expensive: \$5-400MM



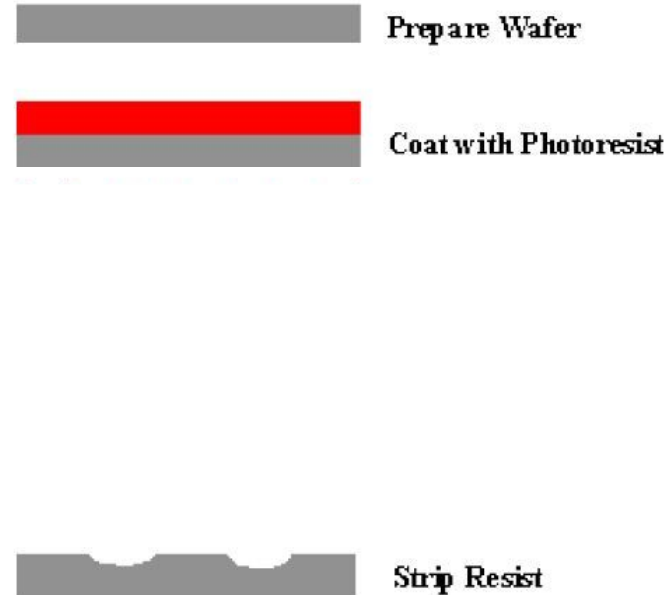
Prepare Wafer

How do we go from top to bottom?

Strip Resist

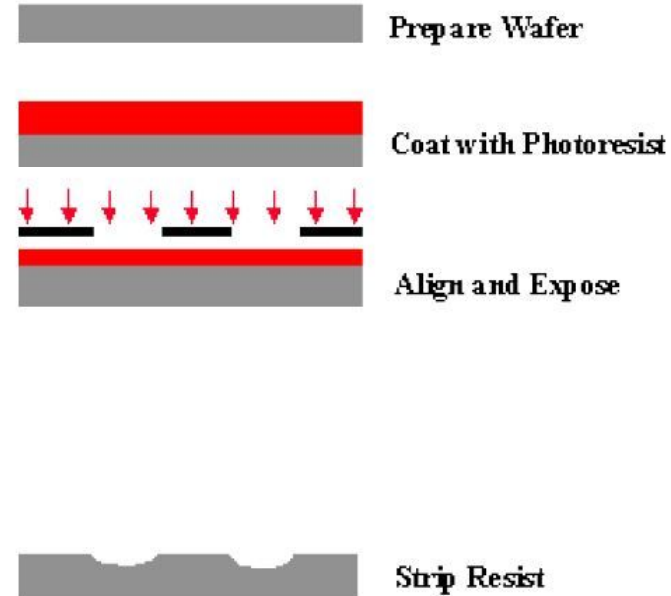
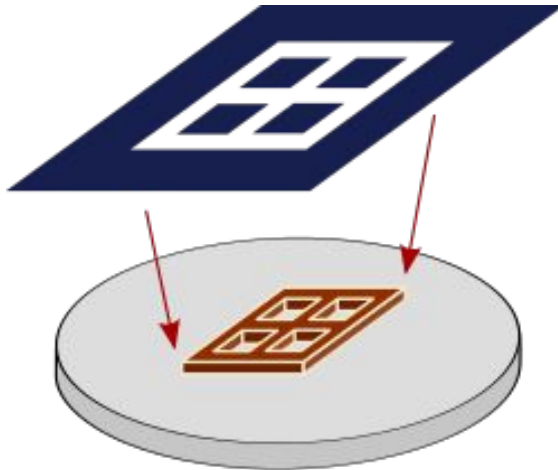
Lithography: Photoresist (PR)

- Viscous liquid that sticks to surface of the wafer
- Reacts when exposed to certain wavelength of light



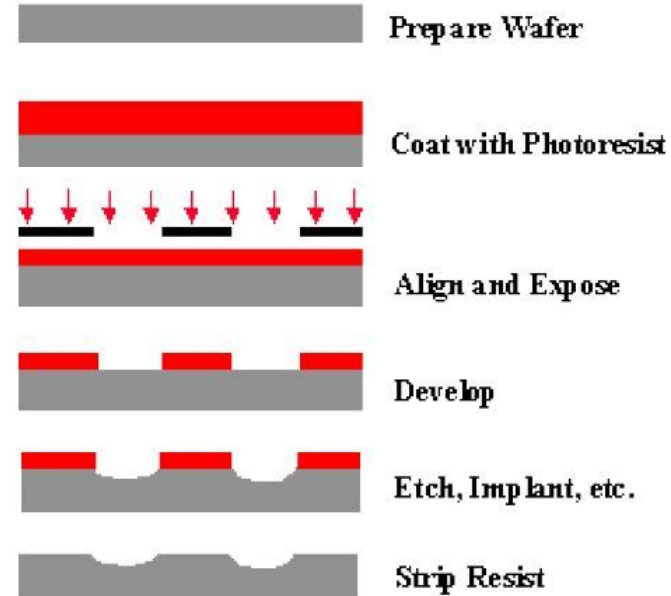
Lithography: Photomask and Exposure

- Photomask selectively allows light through to expose certain regions of PR
- PR becomes more soluble when exposed



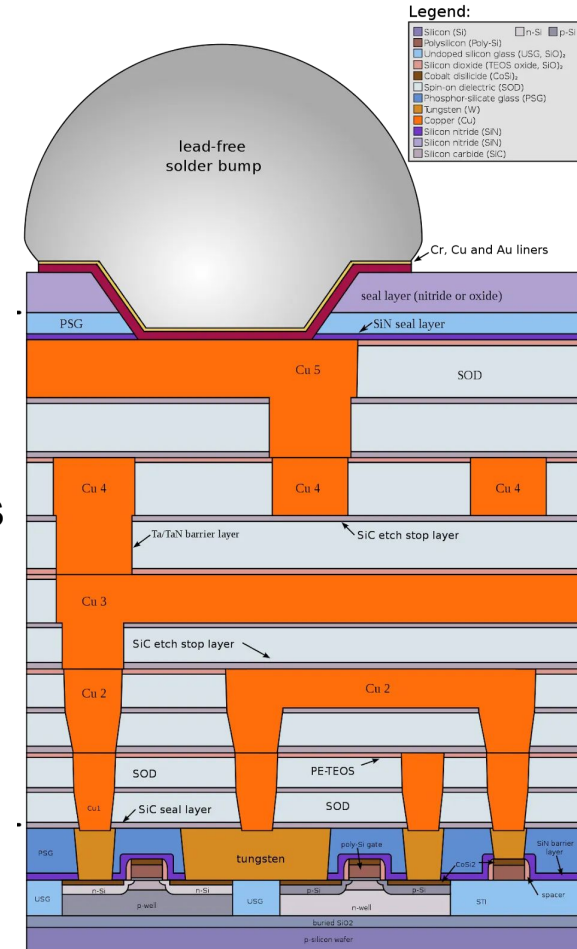
Lithography: Development and Strip

- PR developer removes exposed regions without affecting unexposed regions
- Allows “windows” into underlying surface for more processing:
 - Etch: Remove exposed material
 - Implant: Shoot ions through the window
 - PR acts as a protective barrier elsewhere



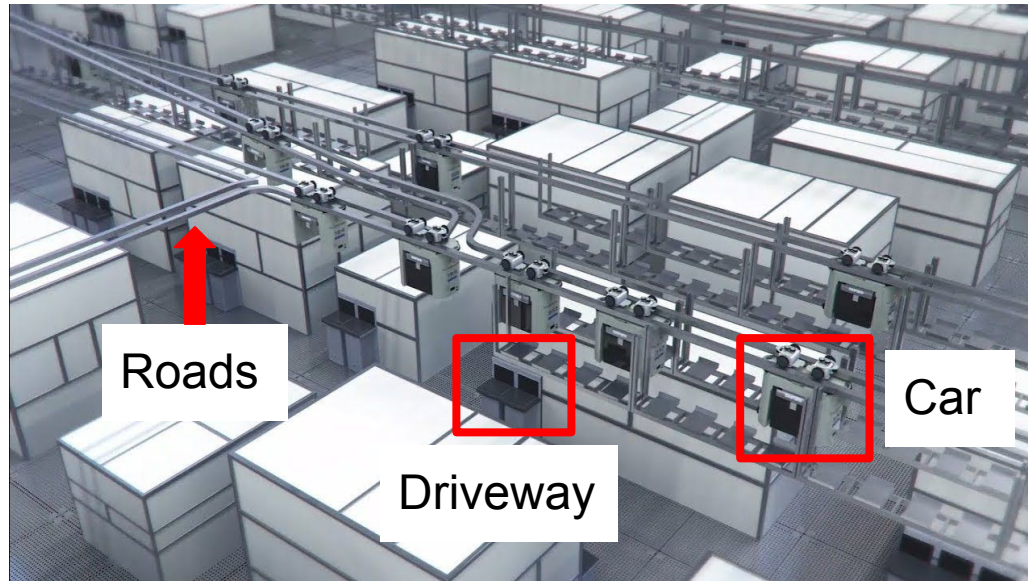
Automation: MES: The Brains of the Fab

- Manufacturing execution system (MES) controls all material in the fab
 - Where it goes
 - What it does
- Crucial given complexity and scale of semi operations
- Automate, damn it!



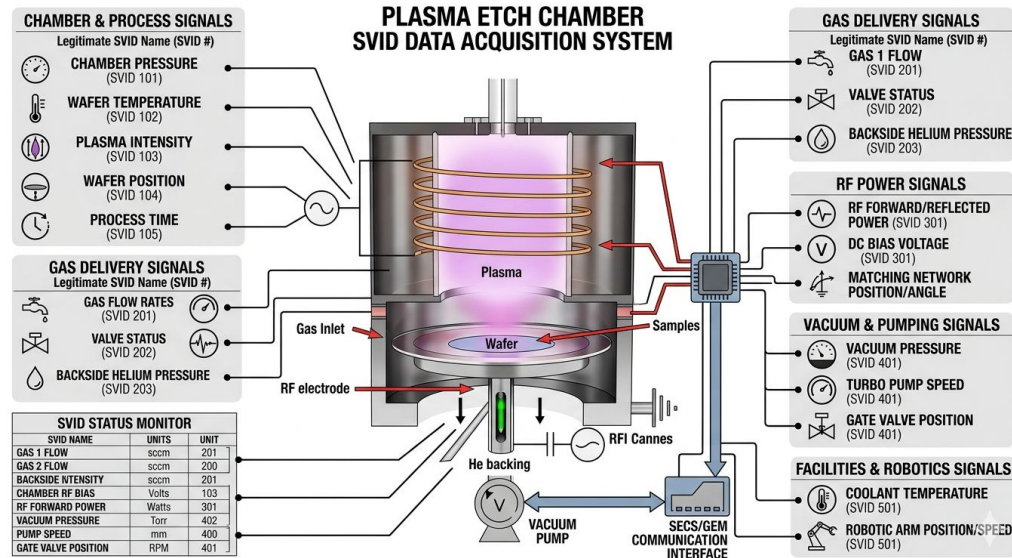
Automation: Automated Material Handling System (AMHS)

- Waymo, but people are wafers and destinations are tools
- 100s of cars along miles of track for safe, fast, reliable pickup and dropoff
- Each tool has multiple “driveways” to park wafers on



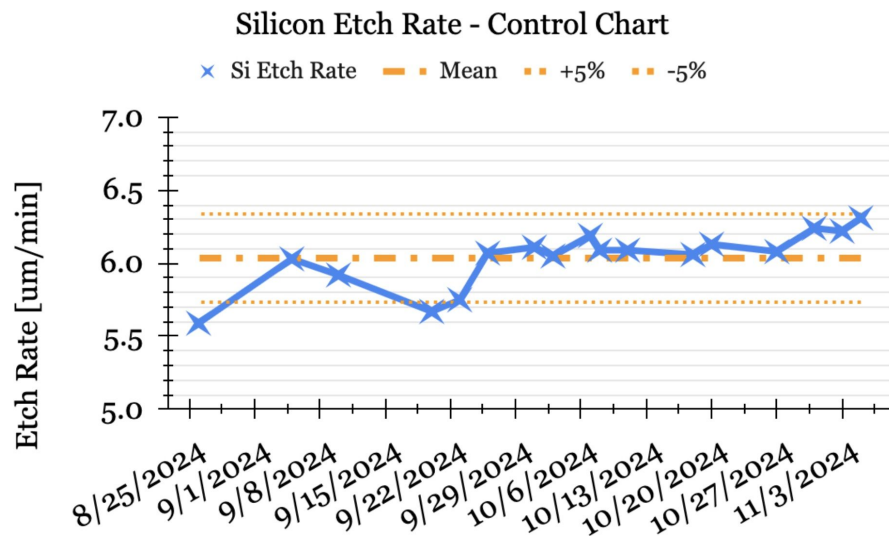
Automation: Using Data to Understand and Interdict

- Data connects patterns, solves problems, improves decision making, etc
- 100s of signals per tool per second = 100k+ signals per second per fab!
- Volume requires automated monitoring and interdiction



Automation: Process Adjustments

- Individual processes must constantly be adjusted to hit targets based on data
- Adjustments can be made at the wafer level, lot level, tool level, etc
- 1000s of daily adjustments requires automation and robust modeling



Operations - Groups

- Process integration: Works with designers to create optimal processes
- Equip eng: Responsible for equip performance and improvement
- Process eng: Responsible for process performance and improvement
- Manufacturing: Oversees day-to-day operations
- Technicians: 24/7/365 coverage; EE and PE fix/maintain tools and processes
- Facilities: Monitors and maintains power/gas/water equipment
- Quality: Ensures standards are being met and protocols followed

Where Do I See Fabs Going?

- Multi-pronged approach by many parties to make everything better:
 - Transistor and chip design: faster, cheaper, more reliable, simpler
 - Equipment: faster, cheaper, more reliable, simpler
 - Process: faster, cheaper, more reliable, simpler
 - Etc etc etc

Slowly converging on utopia!



See Also

- Equipment: nomagicpill.github.io/research/fabs.html
- Operations: nomagicpill.substack.com/p/semiconductor-fabs-ii-the-operation
- Data and Automation: nomagicpill.substack.com/p/semiconductor-fabs-iii-the-data-and
- Ion Implantation: nomagicpill.github.io/research/implant.html
- The Building: construction-physics.com/p/how-to-build-a-20-billion-semiconductor
- Homemade Fab: sam.zeloof.xyz

Questions?